Presence-Filtered Self-Training for Instrument Detection and Template Answering with Tool Priors

Omar Choudhry
Artificial Intelligence in Medicine and Surgery Group, School of Computer Science,
University of Leeds, Leeds LS2 9JT, United Kingdom
O.Choudhry@leeds.ac.uk ORCID 0000-0003-4434-3550
Team AIMS Lab · Categories 1 and 2

1 Introduction

We submitted two independent systems for SurgVU 2026. Our Category 1 detector returns instrument boxes, classes and confidence scores for every video frame. We combine human box annotations with detector proposals filtered by tool-installation labels to train the clevis-scale detector. Our Category 2 answerer combines the question's grammatical form with instrument frequencies from the training labels. It generates short declarative answers without reading the video. Figure 1 shows both inference paths.

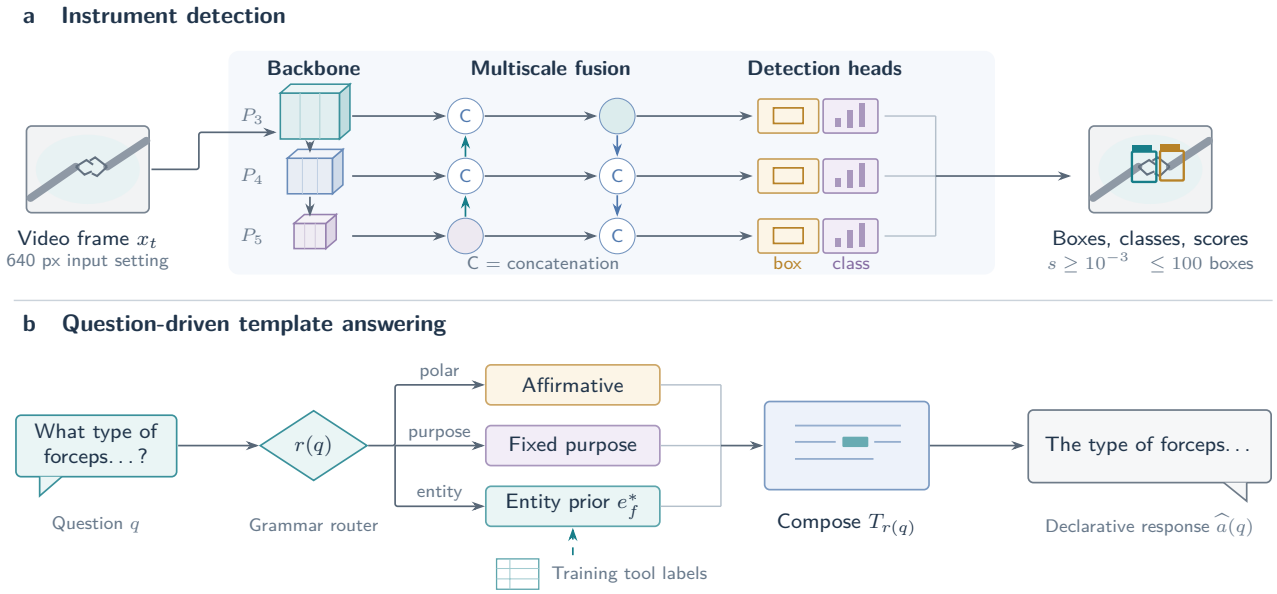


Figure 1: The two inference paths. (a) A framewise YOLO26 detector with multiscale feature fusion. Tensors and instrument frames are schematic. (b) Grammar routes a question to an affirmative polarity, a fixed purpose phrase or an entity prior, then a template composes the response.

2 Category 1

Detector and inference. We initialise a medium YOLO26 model [1] from pretrained weights. At inference, the container loads the selected detector checkpoint and processes every decoded frame independently. With input-size setting 640 pixels, the wrapper retains at most 100 detections above confidence 10^{-3}. For candidate predictions $f_\theta(x_t)$, the output is

$$\widehat{\mathcal{D}}_t = \mathrm{Top}_{100}\big\{(b, c, s) \in f_\theta(x_t) \mid s \geq 10^{-3}\big\}, \tag{1}$$

where b is a box, $c \in \{1, \ldots, 14\}$ is the instrument class and s is its confidence. The wrapper writes four box corners in the original frame coordinates, with the class and frame index. Inference uses no temporal smoothing or ensemble.

Supervision and training. Human COCO annotations supply the target boxes. We use videos 1–5 for local training and videos 6–7 for validation. Following early experiments with Grad-CAM proposals [2], we use a detector trained on human boxes to generate additional labels. A proposal (b, c, s) is retained only when its confidence passes τ and class c is installed at that time

$$\widetilde{\mathcal{D}}_t^{\mathrm{filter}} = \big\{(b, c) \mid (b, c, s) \in f_\phi(x_t),\ s \geq \tau,\ c \in I_t\big\}, \qquad \tau = 0.25, \tag{2}$$

where I_t is the installed-tool set. We retrain on the retained pseudo-labels and human-labelled frames with seeds 0, 1 and 2, then select the seed-2 checkpoint by local validation mAP. The proposer excludes the held-out videos and uses its last checkpoint. Installation constrains the class label but does not establish visibility.

3 Category 2

Question routing and answer composition. A leading auxiliary routes a question to the polar branch. Otherwise, "purpose" or an initial "why" selects the purpose branch. All remaining questions use the entity branch. The response is

$$\widehat{a}(q) = T_{r(q)}\big(q, z(q)\big), \qquad z(q) = \begin{cases} \text{True} & r(q) = \text{polar}, \\ \text{"to grasp and hold tissue"} & r(q) = \text{purpose}, \\ e(q) & r(q) = \text{entity}. \end{cases} \tag{3}$$

Here r is the router and T the sentence template. Polar answers are always affirmative. Templates reuse the question's wording in declarative order without enforcing a fixed word count.

Instrument prior. The prior uses `tools.csv` and deduplicated task segments in `tasks.csv` across 155 training cases, yielding $N = 1{,}850$ segments. A tool belongs to segment j when its installation interval overlaps that segment. Let S_j be this tool set and $\mathcal{V}_f$ the commercial tool names matching instrument family f. The selected entity is

$$e_f^* = \arg\max_{u \in \mathcal{V}_f} \frac{1}{N} \sum_{j=1}^{N} \mathbf{1}\{u \in S_j\}. \tag{4}$$

The longest matching family name in the question selects f. Organ and procedure questions use fixed phrases. Unmatched entity questions fall back to "the tissue". The visual task classifier is disabled. The active path loads no neural model and opens no video.

Evaluation and interpretation. The organiser's evaluator ranks BERTScore-F1 [3], using `roberta-large` with baseline rescaling. For each question it takes the maximum over the reference answers, then averages across questions. Our diagnostic scripts probe constant answers, polarity and sentence length on 11 public examples. These probes do not establish visual understanding. The fixed affirmative branch and entity prior cannot verify events in an unseen clip.

4 Availability

The implementation is available at `https://github.com/omariosc/surgvu-aims` under the MIT licence.

References

[1] Glenn Jocher, Jing Qiu, Mengyu Liu, Shuai Lyu, Fatih Cagatay Akyon, and Muhammet Esat Kalfaoglu. Ultralytics YOLO26: Unified real-time end-to-end vision models. arXiv:2606.03748, 2026. `https://doi.org/10.48550/arXiv.2606.03748`.

[2] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 618–626, 2017. `https://doi.org/10.1109/ICCV.2017.74`.

[3] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations*, 2020. `https://openreview.net/forum?id=SkeHuCVFDr`.